Q&A
THINGS YOU WANTED TO KNOW!

Ques. WHICH POPULAR WIRELESS MODULES OPERATE ON 2.4GHZ FREQUENCY AND ARE EASILY AVAILABLE IN THE MARKET?

Pamarthi Kanakaraja

Ans. There are several modules operating on 2.4Ghz frequency. Some popular ones are given below.

Xbee module. XBee modules are wireless communication modules based on Zigbee standard. These utilise IEEE 802.15.4 protocol. Zigbee standards range between Bluetooth and Wi-Fi. These are basically radio frequency (RF) modules that can communicate up to 30.5m (100-feet) indoors or 91.4m (300-feet) outdoors. XBee modules allow reliable and basic communication between microcontrollers (MCUs), personal computers and systems, and support point-to-point and multi-point networks.

nRF24L01+ module. This transceiver module uses the newest 2.4GHz transceiver from Nordic Semiconductor, nRF24L01+. The transceiver IC operates in 2.4GHz band and has many features. To design a radio system with nRF24L01+, you need an MCU and a few external passive components. You can operate and configure nRF24L01+ through a serial peripheral interface.

ESP8266 module. It is one of the most popular Wi-Fi modules used in Internet of Things (IoT) projects. It is a self-contained Wi-Fi networking module with integrated TCP/IP protocol stack that can give any MCU access to your Wi-Fi network. It is a low-cost Wi-Fi microchip produced by Espressif Systems. It can host applications or offload Wi-Fi networking functions from another application processor. The module integrates antenna switches, RF baluns, power amplifiers, filters and power management modules, among others.

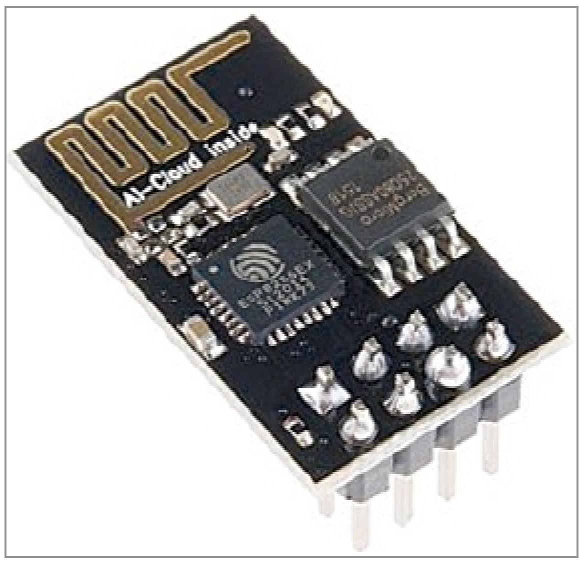

Fig. 1: ESP-01 module

ESP8266-based ESP-01 module operating on 2.4GHz is shown in Fig. 1.

HC-05 Bluetooth module. It is an easy-to-use Bluetooth serial port protocol module, designed for transparent wireless serial connection setups. It is a fully-qualified Bluetooth V2.0+ EDR (enhanced data rate) 3Mbps modulation with complete 2.4GHz radio transceiver and baseband.

TiWi-BLE module. TiWi-BLE module features high-performance 2.4GHz IEEE 802.11 b/g/n, Bluetooth 2.1+EDR and Bluetooth Low Energy 4.0 radio in a cost-effective footprint. The module realises necessary PHY/MAC layers to support WLAN applications in conjunction with a host processor over SDIO interface. The module also provides a Bluetooth platform through HCI transport layer. Both WLAN and Bluetooth share the same antenna port.

Q2. WHAT IS SPINTRONICS, AND WHAT IS ITS SCOPE?

Ashish Mulajkar

A2. Spintronics is a new and exciting field of research in solid-state physics. It is the study of the intrinsic spin of electrons and its associated magnetic moments, apart from the fundamental electronic charge. Magnetic degree of freedom of an electron or its spin is studied and exploited for classical and quantum information processing. One of the objectives is to understand the fundamentals of electron spin and develop new materials based on charge-based electronics with spin-dependent effects.

Spintronics holds promise for future to build efficient and faster computers, new materials and other devices. It has several advantages over conventional electronics. The scope of spintronics is given below.

Faster computing. As the technology advances, the desire to build smaller, faster, cheaper electronics has prompted many researchers to use the spin of an electron in transistors. Spin transistors can be highly energy efficient and do more computation than traditional transistors in a smaller space.

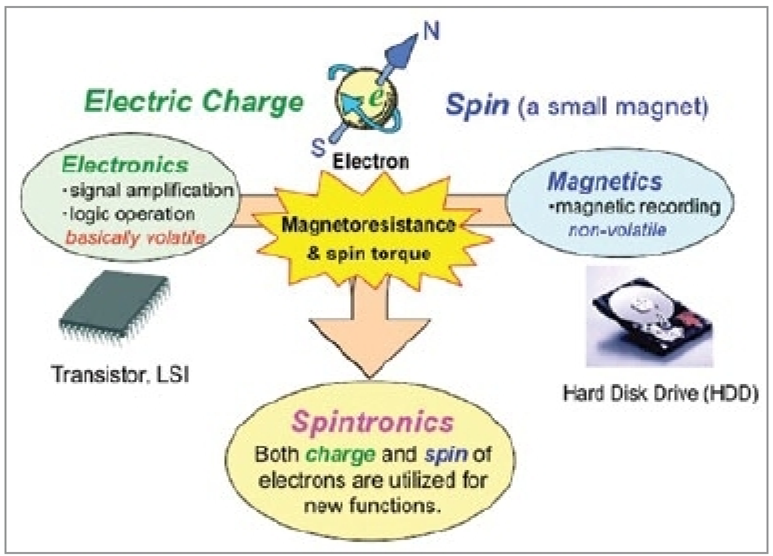

Fig. 2: Concept of spintronics (Credit: unit.aist.go.jp)

Increase data-carrying capacity. In optoelectronic applications, lasers and LEDs that take advantage of the spin of electrons could increase the data-carrying capacity of light.

Less power. Unlike conventional flow of electrical current, spin can be maintained even if power is off. With spintronics, the circuit uses less power because there is no need to apply current constantly.

More storage capacity. Spintronics may help increase information-storing and transmitting capacities of electrons.

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